University Of Mumbai

T.Y. B.Sc. Sem VI (Computer Science) Rev 21 Practical Examination – April 2025

Duration: 2 hrs **Marks: 50**

Date: 15 April 2025 **Time: 9:00 - 11:00**

Course: USCSP601 Data Science – Practical

Candidate's University Seat Number: _____

1.	Implement Decision Tree Model using Iris dataset using Python/R and interpret decision rules of classification.	40
	Viva	5
	Journal	5

SOLUTION – Chit 1

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

iris = load_iris()

X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = DecisionTreeClassifier()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)

print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))

print("\nConfusion Matrix:\n")
print(confusion_matrix(y_test, y_pred))

plt.figure(figsize=(12,8))
plot_tree(model, feature_names=iris.feature_names, class_names=iris.target_names, filled=True)
plt.show()
```

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Duration: 2 hrs **Marks: 50**

Date: 15 April 2025 **Time: 9:00 - 11:00**

Course: USCSP601 Data Science – Practical

Candidate's University Seat Number: _____

1.	Load Iris Dataset. Apply K-means Algorithm using Python/R to group similar data points into clusters. Determine optimal number of clusters using Elbow Method. Visualize clustering results and analyze cluster characteristics.	40
	Viva	5
	Journal	5

SOLUTION – Chit 2

```
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

iris = load_iris()
X = iris.data

inertias = []
for k in range(1, 11):
    km = KMeans(n_clusters=k, random_state=42, n_init=10)
    km.fit(X)
    inertias.append(km.inertia_)

plt.plot(range(1, 11), inertias, marker='o')
plt.title('Elbow Method')
plt.xlabel('Number of Clusters')
plt.ylabel('Inertia')
plt.savefig('elbow.png')
plt.show()

km = KMeans(n_clusters=3, random_state=42, n_init=10)
labels = km.fit_predict(X)

plt.scatter(X[:, 0], X[:, 1], c=labels, cmap='viridis')
plt.title('K-Means Clustering (k=3)')
plt.xlabel('Sepal Length')
plt.ylabel('Sepal Width')
plt.savefig('clusters.png')
plt.show()
```

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Duration: 2 hrs **Marks: 50**

Date: 15 April 2025 **Time: 9:00 - 11:00**

Course: USCSP601 Data Science – Practical

Candidate's University Seat Number: _____

1.	Load Iris Dataset. Apply K-means Algorithm using Python/R to group similar data points into clusters. Determine optimal number of clusters using Silhouette analysis. Visualize clustering results and analyze cluster characteristics.	40
	Viva	5
	Journal	5

SOLUTION – Chit 3

```
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
import matplotlib.pyplot as plt

iris = load_iris()
X = iris.data

scores = []
for k in range(2, 11):
    km = KMeans(n_clusters=k, random_state=42, n_init=10)
    labels = km.fit_predict(X)
    scores.append(silhouette_score(X, labels))

plt.plot(range(2, 11), scores, marker='o')
plt.title('Silhouette Analysis')
plt.xlabel('Number of Clusters')
plt.ylabel('Silhouette Score')
plt.savefig('silhouette.png')
plt.show()

best_k = range(2, 11)[scores.index(max(scores))]
print("Best k:", best_k)

km = KMeans(n_clusters=best_k, random_state=42, n_init=10)
labels = km.fit_predict(X)

plt.scatter(X[:, 0], X[:, 1], c=labels, cmap='viridis')
plt.title(f'K-Means Clustering (k={best_k})')
plt.savefig('clusters.png')
plt.show()
```

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Date: 15 April 2025 **Time: 9:00 - 11:00**

Course: USCSP601 Data Science – Practical

Candidate's University Seat Number: _____

1.	A. Consider a scenario where you have test scores from a sample of students and you want to compare the mean of these scores with hypothesized population mean. Student Score = [72, 88, 64, 74, 67, 79, 85, 75, 89, 77] Apply One Sampled T-Test using Python/R for above problem. Assume hypothesized mean as 70. Formulate Null and Alternative Hypothesis for a given problem. Interpret the results and draw the conclusion. [20] B. Apply Feature Scaling technique like standardization and normalization using Python/R to Boston Housing dataset. [20]	40
	Viva	5
	Journal	5

SOLUTION – Chit 4

Part A – One Sample T-Test

```
from scipy import stats

scores = [72, 88, 64, 74, 67, 79, 85, 75, 89, 77]
t_stat, p_value = stats.ttest_1samp(scores, 70)
print("H0: Mean = 70")
print("H1: Mean != 70")
print(f"T-statistic: {t_stat:.4f}")
print(f"P-value: {p_value:.4f}")
if p_value < 0.05:
    print("Reject H0: Significant difference from mean 70")
else:
    print("Fail to reject H0: No significant difference from mean 70")
```

Part B – Feature Scaling on Boston Housing Dataset

```
import pandas as pd
from sklearn.preprocessing import StandardScaler, MinMaxScaler
from sklearn.datasets import fetch_california_housing

data = fetch_california_housing()
df = pd.DataFrame(data.data, columns=data.feature_names)

scaler_std = StandardScaler()
df_std = pd.DataFrame(scaler_std.fit_transform(df), columns=df.columns)
print("Standardized Data:\n", df_std.head())

scaler_norm = MinMaxScaler()
df_norm = pd.DataFrame(scaler_norm.fit_transform(df), columns=df.columns)
print("Normalized Data:\n", df_norm.head())
```

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Course: USCSP601 Data Science – Practical

Candidate's University Seat Number: _____

1.	A. The employee's aptitude and job proficiency score is as follows. aptitude: 85 65 50 68 87 74 65 96 68 94 73 84 85 87 91 jobprof: 70 90 80 89 88 86 78 67 86 90 92 94 99 93 87 Perform Chi-Square Test to study correlation between aptitude and job proficiency of employee. Formulate Null and Alternative Hypothesis. Interpret results and draw conclusions based on test outcome. [20] B. Perform Logistic Regression on the Iris dataset using Python/R to predict binary outcome. Evaluate model's performance using classification metrics. [20]	40
	Viva	5
	Journal	5

SOLUTION – Chit 5

Part A – Chi-Square Test

```
import numpy as np
from scipy.stats import chi2_contingency

aptitude = [85,65,50,68,87,74,65,96,68,94,73,84,85,87,91]
jobprof  = [70,90,80,89,88,86,78,67,86,90,92,94,99,93,87]

apt_g = ['High' if x >= 75 else 'Low' for x in aptitude]
job_g = ['High' if x >= 85 else 'Low' for x in jobprof]

ct = np.array([
    [sum(1 for a,j in zip(apt_g,job_g) if a=='High' and j=='High'),
     sum(1 for a,j in zip(apt_g,job_g) if a=='High' and j=='Low')],
    [sum(1 for a,j in zip(apt_g,job_g) if a=='Low' and j=='High'),
     sum(1 for a,j in zip(apt_g,job_g) if a=='Low' and j=='Low')]
])
chi2, p, dof, expected = chi2_contingency(ct)
print("H0: No association between aptitude and job proficiency")
print("H1: There is an association")
print(f"Chi2: {chi2:.4f}, p-value: {p:.4f}, dof: {dof}")
print("Reject H0" if p < 0.05 else "Fail to reject H0")
```

Part B – Logistic Regression on Iris

```
from sklearn.datasets import load_iris
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix

iris = load_iris()
X = iris.data[iris.target != 2]
y = iris.target[iris.target != 2]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
clf = LogisticRegression(max_iter=200)
clf.fit(X_train, y_train)
y_pred = clf.predict(X_test)
print(classification_report(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

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Course: USCSP601 Data Science – Practical

Candidate's University Seat Number: _____

1. A. Create CSV file from given data. Read the data from CSV files into a data frame. Perform Data pre-processing tasks such as handling missing values and outliers using Python/R [20] — 40

Country	Age	Salary	Purchased
France	44	72000	No
Spain	27	48000	Yes
Germany	30	54000	No
Spain	38	61000	No
Germany	40		Yes
France	35	58000	Yes
Spain		52000	No
France	48	79000	Yes
Germany	50	83000	No
France	37	67000	Yes

B. Implement Multiple Linear Regression using Python/R on the below housing dataset to predict price of a house. Evaluate model's performance using classification metrics. [20]

Bedrooms	Bathrooms	Sqft_living	Floors	Grade	Sqft_above	Sqft_basement	Price
3	1	1180	1	7	1180	0	221900
3	2.25	2570	2	7	2170	400	538000
2	1	770	1	6	770	0	180000
4	3	1960	1	7	1050	910	604000
3	2	1680	1	8	1680	0	510000
4	4.5	5420	1	11	3890	1530	267800
3	2.25	1715	2	7	1715	0	257500
3	1.5	1060	1	7	1060	0	291850
3	1	1780	1	7	1050	730	229500
3	2.5	1890	2	7	1890	0	323000
3	2.5	3560	1	8	1860	1700	662500

	Viva	5
	Journal	5

SOLUTION – Chit 6

Part A – Data Preprocessing

```
import pandas as pd
import numpy as np

data = {
    'Country':
['France','Spain','Germany','Spain','Germany','France','Spain','France','Germany','France'],
    'Age': [44,27,30,38,40,35,None,48,50,37],
    'Salary': [72000,48000,54000,61000,None,58000,52000,79000,83000,67000],
    'Purchased': ['No','Yes','No','No','Yes','Yes','No','Yes','No','Yes']
}

df = pd.DataFrame(data)
df.to_csv('data.csv', index=False)
df = pd.read_csv('data.csv')

# Fixed: direct assignment instead of inplace=True
df['Age'] = df['Age'].fillna(df['Age'].mean())
df['Salary'] = df['Salary'].fillna(df['Salary'].mean())

# Outlier removal using IQR
Q1 = df['Salary'].quantile(0.25)
Q3 = df['Salary'].quantile(0.75)
IQR = Q3 - Q1
df = df[(df['Salary'] >= Q1 - 1.5*IQR) & (df['Salary'] <= Q3 + 1.5*IQR)]

print(df)
```

Part B – Multiple Linear Regression

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score

housing = {
    'Bedrooms':[3,3,2,4,3,4,3,3,3,3,3],
    'Bathrooms':[1,2.25,1,3,2,4.5,2.25,1.5,1,2.5,2.5],
    'Sqft_living':[1180,2570,770,1960,1680,5420,1715,1060,1780,1890,3560],
    'Price':[221900,538000,180000,604000,510000,267800,257500,291850,229500,323000,662500]
}
df = pd.DataFrame(housing)
X = df.drop('Price', axis=1)
y = df['Price']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("MSE:", mean_squared_error(y_test, y_pred))
print("R2:", r2_score(y_test, y_pred))
```

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Course: USCSP601 Data Science – Practical

Candidate's University Seat Number: _____

1.	A. Apply Feature Scaling technique like standardization and normalization using Python/R to numerical features of below dataset. [20]	40

Make	Model	Color	Mileage	Sell Price
Honda	Accord	Red	63,512	4000
Honda	Accord	Blue	95,135	2500
Toyota	Camry	Black	75,006	45000
Nissan	Altima	Green	69,847	3826
Toyota	Corolla	Black	87,278	2224
Honda	Civic	White	1,38,789	2723
Ford	F-150	Black	89,073	3950
Chevrolet	Silverado	Green	1,09,231	4959
Chevrolet	Impala	Silver	87,675	3791
Dodge	Charger	Silver	34,853	4349
Dodge	Charger	Silver	58,173	4252

B. Implement Multiple Linear Regression on the "Pima Indian Diabetes dataset" using Python/R. [20]

	Viva	5
	Journal	5

SOLUTION – Chit 7

Part A – Feature Scaling

```
import pandas as pd
from sklearn.preprocessing import StandardScaler, MinMaxScaler

data = {

'Make':['Honda','Honda','Toyota','Nissan','Toyota','Honda','Ford','Chevrolet','Chevrolet','Dodge','
Dodge'],
    'Mileage':[63512,95135,75006,69847,87278,138789,89073,109231,87675,34853,58173],
    'Sell_Price':[4000,2500,45000,3826,2224,2723,3950,4959,3791,4349,4252]
}
df = pd.DataFrame(data)
num_cols = ['Mileage','Sell_Price']

scaler_std = StandardScaler()
df_std = df.copy()
df_std[num_cols] = scaler_std.fit_transform(df[num_cols])
print("Standardized:\n", df_std[num_cols])

scaler_norm = MinMaxScaler()
df_norm = df.copy()
df_norm[num_cols] = scaler_norm.fit_transform(df[num_cols])
print("Normalized:\n", df_norm[num_cols])
```

Part B – Multiple Linear Regression on Pima Diabetes

```
from sklearn.datasets import fetch_openml
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score

diabetes = fetch_openml(name='diabetes', version=1, as_frame=True)
X = diabetes.data
y = diabetes.target.astype(float)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("MSE:", mean_squared_error(y_test, y_pred))
print("R2:", r2_score(y_test, y_pred))
```

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Candidate's University Seat Number: _____

1.	A. Consider the dataset given below. Create Pivot Table in Excel for following analysis and visualize the data using PivotChart: [20] ● How many cars do you have by make and model and by color? ● Find out profit margin of all different Make of cars. ● Find out average cost of vehicles. ● Find out percentage of cars of each color.	40
	Viva	5
	Journal	5

Make	Model	Color	Mileage	Sell Price	Buy Price
Honda	Accord	Red	63,512	4000	3000
Honda	Accord	Blue	95,135	2500	2000
Toyota	Camry	Black	75,006	45000	44000
Nissan	Altima	Green	69,847	3826	3000
Toyota	Corolla	Black	87,278	2224	2100
Honda	Civic	White	1,38,789	2723	1900
Ford	F-150	Black	89,073	3950	3000
Chevrolet	Silverado	Green	1,09,231	4959	4500
Chevrolet	Impala	Silver	87,675	3791	3500
Dodge	Charger	Silver	34,853	4349	3500
Dodge	Charger	Silver	58,173	4252	4000

B. Perform Logistic Regression on the Iris dataset using Python/R to predict binary outcome. Evaluate model's performance using classification metrics. [20]

SOLUTION – Chit 8

Part A – Pivot Table Analysis in Python

```
import pandas as pd
import matplotlib.pyplot as plt

data = {

'Make':['Honda','Honda','Toyota','Nissan','Toyota','Honda','Ford','Chevrolet','Chevrolet','Dodge','
Dodge'],
    'Model':['Accord','Accord','Camry','Altima','Corolla','Civic','F-
150','Silverado','Impala','Charger','Charger'],

'Color':['Red','Blue','Black','Green','Black','White','Black','Green','Silver','Silver','Silver'],
    'Mileage':[63512,95135,75006,69847,87278,138789,89073,109231,87675,34853,58173],
    'Sell_Price':[4000,2500,45000,3826,2224,2723,3950,4959,3791,4349,4252],
    'Buy_Price':[3000,2000,44000,3000,2100,1900,3000,4500,3500,3500,4000]
}
df = pd.DataFrame(data)

print(df.groupby(['Make','Model','Color']).size())

df['Profit'] = df['Sell_Price'] - df['Buy_Price']
print(df.groupby('Make')['Profit'].sum())

print("Average cost:", df['Buy_Price'].mean())

color_pct = df['Color'].value_counts(normalize=True) * 100
print(color_pct)
color_pct.plot(kind='pie', autopct='%1.1f%%', title='Cars by Color')
plt.savefig('color_pie.png')
plt.show()
```

Part B – Logistic Regression on Iris

```
from sklearn.datasets import load_iris
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix

iris = load_iris()
X = iris.data[iris.target != 2]
y = iris.target[iris.target != 2]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
clf = LogisticRegression(max_iter=200)
clf.fit(X_train, y_train)
y_pred = clf.predict(X_test)
print(classification_report(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

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<table>
<tr><td>1.</td><td>A. Apply Feature Scaling technique like standardization and normalization using Python/R to numerical features of below dataset. [20]

<table>
<tr><th>Country</th><th>Age</th><th>Salary</th><th>Purchased</th></tr>
<tr><td>France</td><td>44</td><td>72000</td><td>No</td></tr>
<tr><td>Spain</td><td>27</td><td>48000</td><td>Yes</td></tr>
<tr><td>Germany</td><td>30</td><td>54000</td><td>No</td></tr>
<tr><td>Spain</td><td>38</td><td>61000</td><td>No</td></tr>
<tr><td>Germany</td><td>40</td><td>85000</td><td>Yes</td></tr>
<tr><td>France</td><td>35</td><td>58000</td><td>Yes</td></tr>
<tr><td>Spain</td><td>31</td><td>52000</td><td>No</td></tr>
<tr><td>France</td><td>48</td><td>79000</td><td>Yes</td></tr>
<tr><td>Germany</td><td>50</td><td>83000</td><td>No</td></tr>
<tr><td>France</td><td>37</td><td>67000</td><td>Yes</td></tr>
</table>

B. Load the Iris dataset. Perform Principal Component Analysis (PCA) using Python/R on a dataset to reduce dimensionality. Select appropriate number of principle components. Visualize the data in the reduced-dimensional space. [20]</td><td>40</td></tr>
<tr><td></td><td>Viva</td><td>5</td></tr>
<tr><td></td><td>Journal</td><td>5</td></tr>
</table>

SOLUTION – Chit 9

Part A – Feature Scaling

```
import pandas as pd
from sklearn.preprocessing import StandardScaler, MinMaxScaler

data = {

'Country':['France','Spain','Germany','Spain','Germany','France','Spain','France','Germany','France
'],
    'Age':[44,27,30,38,40,35,31,48,50,37],
    'Salary':[72000,48000,54000,61000,85000,58000,52000,79000,83000,67000],
    'Purchased':['No','Yes','No','No','Yes','Yes','No','Yes','No','Yes']
}
df = pd.DataFrame(data)
num_cols = ['Age','Salary']

scaler_std = StandardScaler()
df_std = df.copy()
df_std[num_cols] = scaler_std.fit_transform(df[num_cols])
print("Standardized:\n", df_std[num_cols])

scaler_norm = MinMaxScaler()
df_norm = df.copy()
df_norm[num_cols] = scaler_norm.fit_transform(df[num_cols])
print("Normalized:\n", df_norm[num_cols])
```

Part B – PCA on Iris Dataset

```
from sklearn.datasets import load_iris
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

iris = load_iris()
X, y = iris.data, iris.target

pca = PCA()
pca.fit(X)
cum_var = pca.explained_variance_ratio_.cumsum()
n_comp = next(i+1 for i, v in enumerate(cum_var) if v >= 0.95)
print("Components for 95% variance:", n_comp)

pca2 = PCA(n_components=2)
X_pca = pca2.fit_transform(X)

for label in set(y):
    plt.scatter(X_pca[y==label,0], X_pca[y==label,1], label=iris.target_names[label])
plt.xlabel('PC1')
plt.ylabel('PC2')
plt.title('PCA of Iris Dataset')
plt.legend()
plt.savefig('pca.png')
plt.show()
```

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Candidate's University Seat Number: _____

1.	A. Convert Categorical Column to numerical representation (Feature Dummification) of below dataset using Python/R. [20]	40

Country	Age	Salary	Purchased
France	44	72000	No
Spain	27	48000	Yes
Germany	30	54000	No
Spain	38	61000	No
Germany	40	85000	Yes
France	35	58000	Yes
Spain	31	52000	No
France	48	79000	Yes
Germany	50	83000	No
France	37	67000	Yes

B. Perform Logistic Regression on the Iris dataset using Python/R to predict binary outcome. Evaluate model's performance using classification metrics. [20]

	Viva	5
	Journal	5

SOLUTION – Chit 10

Part A – Feature Dummification

```
import pandas as pd

data = {

'Country':['France','Spain','Germany','Spain','Germany','France','Spain','France','Germany','France
'],
    'Age':[44,27,30,38,40,35,31,48,50,37],
    'Salary':[72000,48000,54000,61000,85000,58000,52000,79000,83000,67000],
    'Purchased':['No','Yes','No','No','Yes','Yes','No','Yes','No','Yes']
}
df = pd.DataFrame(data)
df_encoded = pd.get_dummies(df, columns=['Country','Purchased'], drop_first=True)
print(df_encoded)
```

Part B – Logistic Regression on Iris

```
from sklearn.datasets import load_iris
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix

iris = load_iris()
X = iris.data[iris.target != 2]
y = iris.target[iris.target != 2]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
clf = LogisticRegression(max_iter=200)
clf.fit(X_train, y_train)
y_pred = clf.predict(X_test)
print(classification_report(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

University Of Mumbai

T.Y. B.Sc. Sem VI (Computer Science) Rev 21 Practical Examination – April 2025

Duration: 2 hrs **Marks: 50**

Date: 15 April 2025 **Time: 9:00 - 11:00**

Course: USCSP601 Data Science – Practical

Candidate's University Seat Number: _____

1.	A. Create CSV file from given data. Read the data from CSV files into a data frame. Perform transformation functions on given data using Python/R. [20] ● Display records of the car having Sell Price greater than 4000. ● Sort the car data in ascending order. ● Group the data according the "Make" of car.	40
	Viva	5
	Journal	5

Make	Model	Color	Mileage	Sell Price	Buy Price
Honda	Accord	Red	63,512	4000	3000
Honda	Accord	Blue	95,135	2500	2000
Toyota	Camry	Black	75,006	45000	44000
Nissan	Altima	Green	69,847	3826	3000
Toyota	Corolla	Black	87,278	2224	2100
Honda	Civic	White	1,38,789	2723	1900
Ford	F-150	Black	89,073	3950	3000
Chevrolet	Silverado	Green	1,09,231	4959	4500
Chevrolet	Impala	Silver	87,675	3791	3500
Dodge	Charger	Silver	34,853	4349	3500
Dodge	Charger	Silver	58,173	4252	4000

B. Load the Iris dataset. Perform Principal Component Analysis (PCA) using Python/R on a dataset to reduce dimensionality. Select appropriate number of principle components. Visualize the data in the reduced-dimensional space. [20]

SOLUTION – Chit 11

Part A – Data Transformation

```
import pandas as pd

data = {

'Make':['Honda','Honda','Toyota','Nissan','Toyota','Honda','Ford','Chevrolet','Chevrolet','Dodge','
Dodge'],
    'Model':['Accord','Accord','Camry','Altima','Corolla','Civic','F-
150','Silverado','Impala','Charger','Charger'],

'Color':['Red','Blue','Black','Green','Black','White','Black','Green','Silver','Silver','Silver'],
    'Mileage':[63512,95135,75006,69847,87278,138789,89073,109231,87675,34853,58173],
    'Sell_Price':[4000,2500,45000,3826,2224,2723,3950,4959,3791,4349,4252],
    'Buy_Price':[3000,2000,44000,3000,2100,1900,3000,4500,3500,3500,4000]
}
df = pd.DataFrame(data)
df.to_csv('cars.csv', index=False)
df = pd.read_csv('cars.csv')

print("Sell Price > 4000:\n", df[df['Sell_Price'] > 4000])
print("Sorted by Sell Price:\n", df.sort_values('Sell_Price'))
print("Grouped by Make:\n", df.groupby('Make')['Sell_Price'].mean())
```

Part B – PCA on Iris

```
from sklearn.datasets import load_iris
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

iris = load_iris()
X, y = iris.data, iris.target

pca2 = PCA(n_components=2)
X_pca = pca2.fit_transform(X)
print("Explained variance:", pca2.explained_variance_ratio_)

for label in set(y):
    plt.scatter(X_pca[y==label,0], X_pca[y==label,1], label=iris.target_names[label])
plt.xlabel('PC1')
plt.ylabel('PC2')
plt.title('PCA of Iris Dataset')
plt.legend()
plt.savefig('pca.png')
plt.show()
```

University Of Mumbai

T.Y. B.Sc. Sem VI (Computer Science) Rev 21 Practical Examination – April 2025

Duration: 2 hrs **Marks: 50**

Date: 15 April 2025 **Time: 9:00 - 11:00**

Course: USCSP601 Data Science – Practical

Candidate's University Seat Number: _____

1.	A. Consider the dataset given below. Class A = [85, 90, 88, 82, 87] Class B = [76, 78, 80, 81, 75] Class C = [92, 88, 94, 89, 90] Perform One Way ANOVA Test using Python/R to determine if there is a significant difference in the mean exam scores among these classes. Formulate Null and Alternative Hypothesis for a given problem. Interpret results and draw conclusions based on test outcome. [20] B. Load the Wine Quality dataset. Perform Principal Component Analysis (PCA) using Python/R on a dataset to reduce dimensionality. Select appropriate number of principle components. Visualize the data in the reduced-dimensional space. [20]	40
	Viva	5
	Journal	5

SOLUTION – Chit 12

Part A – One-Way ANOVA

```
from scipy.stats import f_oneway

class_a = [85,90,88,82,87]
class_b = [76,78,80,81,75]
class_c = [92,88,94,89,90]

f_stat, p_value = f_oneway(class_a, class_b, class_c)
print("H0: All class means are equal")
print("H1: At least one class mean is different")
print(f"F-statistic: {f_stat:.4f}")
print(f"P-value: {p_value:.4f}")
print("Reject H0" if p_value < 0.05 else "Fail to reject H0")
```

Part B – PCA on Wine Quality Dataset

```
import pandas as pd
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
import matplotlib.pyplot as plt

df = pd.read_csv('https://archive.ics.uci.edu/ml/machine-learning-databases/wine-
quality/winequality-red.csv', sep=';')
X = df.drop('quality', axis=1)
y = df['quality']

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

pca = PCA()
pca.fit(X_scaled)
cum_var = pca.explained_variance_ratio_.cumsum()
n_comp = next(i+1 for i, v in enumerate(cum_var) if v >= 0.95)
print("Components for 95% variance:", n_comp)

pca2 = PCA(n_components=2)
X_pca = pca2.fit_transform(X_scaled)

plt.scatter(X_pca[:,0], X_pca[:,1], c=y, cmap='viridis', alpha=0.5)
plt.colorbar(label='Quality')
plt.xlabel('PC1')
plt.ylabel('PC2')
plt.title('PCA of Wine Quality Dataset')
plt.savefig('pca_wine.png')
plt.show()
```

University Of Mumbai

T.Y. B.Sc. Sem VI (Computer Science) Rev 21 Practical Examination – April 2025

Duration: 2 hrs **Marks: 50**

Date: 15 April 2025 **Time: 9:00 - 11:00**

Course: USCSP601 Data Science – Practical

Candidate's University Seat Number: _____

1.	A. The company is accessing the difference in time to complete the task between two groups of employees. State the hypothesis and do the Two Sampled T-Test using Python/R for the given dataset. [20] Group-I: Experience(0-1 year) Group-II: Experience(1-2 year) Time taken by Group1: 85,95,100,80,90,97,104,95,88,92,94,99 Time taken by Group2: 83,85,96,92,100,104,94,95,88,90,93,94 B. Perform Multiple Linear Regression on the "Pima Indian Diabetes dataset" using Python/R. [20]	40
	Viva	5
	Journal	5

SOLUTION – Chit 13

Part A – Two Sample T-Test

```
from scipy import stats

group1 = [85,95,100,80,90,97,104,95,88,92,94,99]
group2 = [83,85,96,92,100,104,94,95,88,90,93,94]

t_stat, p_value = stats.ttest_ind(group1, group2)
print("H0: Mean time of Group1 = Mean time of Group2")
print("H1: Mean time of Group1 != Mean time of Group2")
print(f"T-statistic: {t_stat:.4f}")
print(f"P-value: {p_value:.4f}")
print("Reject H0" if p_value < 0.05 else "Fail to reject H0")
```

Part B – Multiple Linear Regression on Pima Diabetes

```
from sklearn.datasets import fetch_openml
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score

diabetes = fetch_openml(name='diabetes', version=1, as_frame=True)
X = diabetes.data
y = diabetes.target.astype(float)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("MSE:", mean_squared_error(y_test, y_pred))
print("R2:", r2_score(y_test, y_pred))
```

University Of Mumbai

T.Y. B.Sc. Sem VI (Computer Science) Rev 21 Practical Examination – April 2025

Duration: 2 hrs **Marks: 50**

Date: 15 April 2025 **Time: 9:00 - 11:00**

Course: USCSP601 Data Science – Practical

Candidate's University Seat Number: _____

1.	Construct a Decision Tree using Python/R to classify whether a person can play tennis or not. Test the model and check prediction of the model is correct or not. The following data set recorded whether or not a person played tennis depending on the outlook and wind conditions. ● Each instance (example) is represented by the three attributes. o Outlook: a value of {Sunny, Overcast, Rain}. o Wind: a value of {Weak, Strong}. o PlayTennis: the classification attribute (i.e., Yes- the person plays tennis; No - the person does not play tennis).	40
	Viva	5
	Journal	5

Date	Outlook	Wind	PlayTennis	Date	Outlook	Wind	PlayTennis
1	Sunny	Weak	No	11	Sunny	Strong	Yes
2	Sunny	Strong	No	12	Overcast	Strong	Yes
3	Overcast	Weak	Yes	13	Overcast	Weak	Yes
4	Rain	Weak	Yes	14	Rain	Strong	No
5	Rain	Weak	Yes	15	Sunny	Strong	Yes
6	Rain	Strong	No	16	Overcast	Strong	No
7	Overcast	Strong	Yes	17	Overcast	Weak	Yes
8	Sunny	Weak	No	18	Rain	Weak	No
9	Sunny	Weak	Yes	19	Sunny	Weak	No
10	Rain	Weak	Yes	20	Rain	Strong	Yes

SOLUTION – Chit 14

```
import pandas as pd
from sklearn.tree import DecisionTreeClassifier, export_text
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score

data = {
    'Outlook':['Sunny','Sunny','Overcast','Rain','Rain','Rain','Overcast','Sunny','Sunny','Rain',

'Sunny','Overcast','Overcast','Rain','Sunny','Overcast','Overcast','Rain','Sunny','Rain'],
    'Wind':['Weak','Strong','Weak','Weak','Weak','Strong','Strong','Weak','Weak','Weak',
            'Strong','Strong','Weak','Strong','Strong','Strong','Weak','Weak','Weak','Strong'],
    'PlayTennis':['No','No','Yes','Yes','Yes','No','Yes','No','Yes','Yes',
                  'Yes','Yes','Yes','No','Yes','No','Yes','No','No','Yes']
}
df = pd.DataFrame(data)

le_outlook = LabelEncoder()
le_wind = LabelEncoder()
le_play = LabelEncoder()

df['Outlook_enc'] = le_outlook.fit_transform(df['Outlook'])
df['Wind_enc'] = le_wind.fit_transform(df['Wind'])
df['Play_enc'] = le_play.fit_transform(df['PlayTennis'])

X = df[['Outlook_enc','Wind_enc']]
y = df['Play_enc']

clf = DecisionTreeClassifier(criterion='entropy', random_state=42)
clf.fit(X, y)

print("Accuracy:", accuracy_score(y, clf.predict(X)))
print(export_text(clf, feature_names=['Outlook','Wind']))

test = pd.DataFrame({
    'Outlook_enc': le_outlook.transform(['Sunny']),
    'Wind_enc':    le_wind.transform(['Weak'])
})
print("Prediction for Sunny/Weak:", le_play.inverse_transform(clf.predict(test)))
```

University Of Mumbai

T.Y. B.Sc. Sem VI (Computer Science) Rev 21 Practical Examination – April 2025

Duration: 2 hrs **Marks: 50**

Date: 15 April 2025 **Time: 9:00 - 11:00**

Course: USCSP601 Data Science – Practical

Candidate's University Seat Number: _____

1. A. Implement Decision Tree Model on Titanic dataset using Python/R and interpret decision rules of classification. Perform Linear Regression on the following dataset in Python/R for predicting the weight of the person depending on height. [20] — 40

Height	**151**	**174**	**138**	**186**	**128**	**136**	**179**	**163**	**152**
Weight	63	81	56	91	47	57	76	72	62

B. Create CSV file from given data. Read the data from CSV files into a data frame.

Perform transformation function on given data using Python/R. [20]

- Display records of the car having Buy Price greater than equal to 3000.
- Sort the car data in ascending order.
- Group the data according to the "Model" of car.

Make	**Model**	**Color**	**Mileage**	**Sell Price**	**Buy Price**
Honda	Accord	Red	63,512	4000	3000
Honda	Accord	Blue	95,135	2500	2000
Toyota	Camry	Black	75,006	45000	44000
Nissan	Altima	Green	69,847	3826	3000
Toyota	Corolla	Black	87,278	2224	2100
Honda	Civic	White	1,38,789	2723	1900
Ford	F-150	Black	89,073	3950	3000
Chevrolet	Silverado	Green	1,09,231	4959	4500
Chevrolet	Impala	Silver	87,675	3791	3500
Dodge	Charger	Silver	34,853	4349	3500
Dodge	Charger	Silver	58,173	4252	4000

	Viva	5
	Journal	5

SOLUTION – Chit 15

Part A – Decision Tree on Titanic + Linear Regression

```
import pandas as pd
import numpy as np
from sklearn.tree import DecisionTreeClassifier, export_text
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, r2_score

titanic =
pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')
titanic.dropna(subset=['Age'], inplace=True)
titanic['Sex'] = titanic['Sex'].map({'male':0,'female':1})
X = titanic[['Pclass','Sex','Age','Fare']]
y = titanic['Survived']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
clf = DecisionTreeClassifier(max_depth=3, random_state=42)
clf.fit(X_train, y_train)
print("Titanic Accuracy:", accuracy_score(y_test, clf.predict(X_test)))
print(export_text(clf, feature_names=list(X.columns)))

height = np.array([151,174,138,186,128,136,179,163,152]).reshape(-1,1)
weight = np.array([63,81,56,91,47,57,76,72,62])
reg = LinearRegression()
reg.fit(height, weight)
print("R2:", r2_score(weight, reg.predict(height)))
```

Part B – CSV Transformation

```
import pandas as pd

data = {

'Make':['Honda','Honda','Toyota','Nissan','Toyota','Honda','Ford','Chevrolet','Chevrolet','Dodge','
Dodge'],
    'Model':['Accord','Accord','Camry','Altima','Corolla','Civic','F-
150','Silverado','Impala','Charger','Charger'],

'Color':['Red','Blue','Black','Green','Black','White','Black','Green','Silver','Silver','Silver'],
    'Mileage':[63512,95135,75006,69847,87278,138789,89073,109231,87675,34853,58173],
    'Sell_Price':[4000,2500,45000,3826,2224,2723,3950,4959,3791,4349,4252],
    'Buy_Price':[3000,2000,44000,3000,2100,1900,3000,4500,3500,3500,4000]
}
df = pd.DataFrame(data)
df.to_csv('cars.csv', index=False)
df = pd.read_csv('cars.csv')

print("Buy Price >= 3000:\n", df[df['Buy_Price'] >= 3000])
print("Sorted:\n", df.sort_values('Sell_Price'))
print("Grouped by Model:\n", df.groupby('Model')['Sell_Price'].mean())
```

University Of Mumbai

T.Y. B.Sc. Sem VI (Computer Science) Rev 21 Practical Examination – April 2025

Duration: 2 hrs **Marks: 50**

Date: 15 April 2025 **Time: 9:00 - 11:00**

Course: USCSP601 Data Science – Practical

Candidate's University Seat Number: _____

<table>
<tr><td>1.</td><td>A. Perform Linear Regression on the following dataset in Python/R for predicting the salary of the person depending on his/her years of experience. [20]

<table>
<tr><th>Years of Experience</th><th>Salary</th></tr>
<tr><td>2</td><td>30000</td></tr>
<tr><td>10</td><td>95000</td></tr>
<tr><td>4</td><td>45000</td></tr>
<tr><td>20</td><td>178000</td></tr>
<tr><td>8</td><td>84000</td></tr>
<tr><td>12</td><td>120000</td></tr>
<tr><td>22</td><td>200000</td></tr>
</table>

B. Consider dataset given below. Apply VLOOKUP function to retrieve information for the following queries. Also write down the steps for the same. [20]
● Find the Part Name for Part Number "A002".
● Find the Supplier ID for the Part Name "Ball Joint".
● Find the Part Price for Part Name "muffler".
● Find the Status of Part Number "A008".</td><td>40</td></tr>
<tr><td></td><td>Viva</td><td>5</td></tr>
<tr><td></td><td>Journal</td><td>5</td></tr>
</table>

SOLUTION – Chit 16

Part A – Linear Regression (Experience vs Salary)

```
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
import matplotlib.pyplot as plt

years = np.array([2,10,4,20,8,12,22]).reshape(-1,1)
salary = np.array([30000,95000,45000,178000,84000,120000,200000])

model = LinearRegression()
model.fit(years, salary)
y_pred = model.predict(years)
print("Coefficient:", model.coef_[0])
print("Intercept:", model.intercept_)
print("R2:", r2_score(salary, y_pred))

plt.scatter(years, salary, color='blue', label='Actual')
plt.plot(years, y_pred, color='red', label='Predicted')
plt.xlabel('Years of Experience')
plt.ylabel('Salary')
plt.title('Linear Regression: Experience vs Salary')
plt.legend()
plt.savefig('salary_regression.png')
plt.show()
```

Part B – VLOOKUP using Pandas

```
import pandas as pd

data = {

'Supplier_ID':['SP301','SP302','SP303','SP304','SP305','SP306','SP307','SP308','SP309','SP318'],
    'Part_Number':['A001','A002','A003','A004','A005','A006','A007','A008','A009','A018'],
    'Part_Name':['water','altenator','air filter','wheel bearing','muffler','oil pan','brake
pads','rotors','headlight','Ball Joint'],
    'Part_Price':[6800,3800,4500,3582,1600,1005,6500,8549,6500,2500],
    'Status':['In','In','In','In stock','In','Out of stock','In','Out of stock','In','In']
}
df = pd.DataFrame(data)

print("Part Name for A002:", df.loc[df['Part_Number']=='A002','Part_Name'].values[0])
print("Supplier ID for Ball Joint:", df.loc[df['Part_Name']=='Ball Joint','Supplier_ID'].values[0])
print("Part Price for muffler:", df.loc[df['Part_Name']=='muffler','Part_Price'].values[0])
print("Status for A008:", df.loc[df['Part_Number']=='A008','Status'].values[0])
```

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Course: USCSP601 Data Science – Practical

Candidate's University Seat Number: _____

<table>
<tr><td>1.</td><td>A. Perform Linear Regression on the Iris dataset of R/Python for predicting petal.width on petal.length. [20]

B. A student is enrolled in the class. His/Her current grade is 65 which is average of all type of work during the semester and he/she needs at least 72 to pass this class. Final exam is not yet completed. Apply what-if analysis using Goal seek to determine the marks need to be obtained in Final Exam to complete the goal. [20]

<table>
<tr><th>Semester Work</th><th>Grade</th></tr>
<tr><td>Paper Presentation</td><td>58</td></tr>
<tr><td>Case Study</td><td>70</td></tr>
<tr><td>Assignment</td><td>72</td></tr>
<tr><td>Practical</td><td>60</td></tr>
<tr><td>Final Exam</td><td>60</td></tr>
<tr><td>Final Grade</td><td>65</td></tr>
</table></td><td>40</td></tr>
<tr><td></td><td>Viva</td><td>5</td></tr>
<tr><td></td><td>Journal</td><td>5</td></tr>
</table>

SOLUTION – Chit 17

Part A – Linear Regression on Iris (petal.width ~ petal.length)

```
from sklearn.datasets import load_iris
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
import matplotlib.pyplot as plt

iris = load_iris()
import pandas as pd
df = pd.DataFrame(iris.data, columns=iris.feature_names)

X = df[['petal length (cm)']].values
y = df['petal width (cm)'].values

model = LinearRegression()
model.fit(X, y)
y_pred = model.predict(X)
print("Coefficient:", model.coef_[0])
print("Intercept:", model.intercept_)
print("R2:", r2_score(y, y_pred))

plt.scatter(X, y, alpha=0.5, label='Actual')
plt.plot(X, y_pred, color='red', label='Predicted')
plt.xlabel('Petal Length')
plt.ylabel('Petal Width')
plt.title('Linear Regression on Iris')
plt.legend()
plt.savefig('petal_regression.png')
plt.show()
```

Part B – Goal Seek (What-If Analysis)

```
from scipy.optimize import brentq

known_grades = [58, 70, 72, 60]
target_avg = 72

def avg_with_final(final_exam):
    all_grades = known_grades + [final_exam]
    return sum(all_grades) / len(all_grades) - target_avg

required_final = brentq(avg_with_final, 0, 100)
print(f"Required Final Exam marks to achieve grade {target_avg}: {required_final:.2f}")
```

University Of Mumbai

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Course: USCSP601 Data Science – Practical

Candidate's University Seat Number: _____

1.	A. Perform Linear Regression on the Iris dataset of R/Python for predicting petal.width on petal.length. [20] B. Consider the dataset given below. Apply VLOOKUP function to retrieve information for the following queries. Also write down the steps for the same. [20] 1. Find the Part Name for Part Number "A016". 2. Find the Supplier ID for the Part Name "Oil Pump". 3. Find the Part Price for Part Name "Brake". 4. Find the Status of Part Number "A020".	40
	Viva	5
	Journal	5

SOLUTION – Chit 18

Part A – Linear Regression on Iris

```
from sklearn.datasets import load_iris
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
import pandas as pd
import matplotlib.pyplot as plt

iris = load_iris()
df = pd.DataFrame(iris.data, columns=iris.feature_names)

X = df[['petal length (cm)']].values
y = df['petal width (cm)'].values

model = LinearRegression()
model.fit(X, y)
y_pred = model.predict(X)
print("Coefficient:", model.coef_[0])
print("Intercept:", model.intercept_)
print("R2:", r2_score(y, y_pred))

plt.scatter(X, y, alpha=0.5, label='Actual')
plt.plot(X, y_pred, color='red', label='Predicted')
plt.xlabel('Petal Length')
plt.ylabel('Petal Width')
plt.title('Linear Regression on Iris')
plt.legend()
plt.savefig('iris_regression.png')
plt.show()
```

Part B – VLOOKUP using Pandas

```
import pandas as pd

data = {
    'Supplier_ID':['SP301','SP302','SP303','SP304','SP305','SP306','SP307','SP308','SP309',

'SP310','SP311','SP312','SP313','SP314','SP315','SP316','SP317','SP318','SP319','SP320'],
    'Part_Number':['A001','A002','A003','A004','A005','A006','A007','A008','A009',
                   'A010','A011','A012','A013','A014','A015','A016','A017','A018','A019','A020'],
    'Part_Name':['water','altenator','air filter','wheel bearing','muffler','oil pan','brake
pads','rotors',
                 'headlight','brake','Strut','Deive','CV Boot Kit','Oil Pump','Oil Filter','Fuel
Filter',
                 'Tie Rod End','Ball Joint','Steering Rack','Piston'],

'Part_Price':[6800,3800,4500,3582,1600,1005,6500,8549,6500,1500,4500,1580,2650,4660,4350,1280,1800,
2500,2700,4500],
    'Status':['In','In','In','In','In','Out of stock','In','Out of stock','In','In','In','In',
              'In stock','In','In','In','In stock','In','Out of stock','Out of stock']
}
df = pd.DataFrame(data)

print("Part Name for A016:", df.loc[df['Part_Number']=='A016','Part_Name'].values[0])
print("Supplier ID for Oil Pump:", df.loc[df['Part_Name']=='Oil Pump','Supplier_ID'].values[0])
print("Part Price for Brake:", df.loc[df['Part_Name']=='brake','Part_Price'].values[0])
print("Status for A020:", df.loc[df['Part_Number']=='A020','Status'].values[0])
```

University Of Mumbai

T.Y. B.Sc. Sem VI (Computer Science) Rev 21 Practical Examination – April 2025

Duration: 2 hrs **Marks: 50**

Date: 15 April 2025 **Time: 9:00 - 11:00**

Course: USCSP601 Data Science – Practical

Candidate's University Seat Number: _____

<table>
<tr><td>1.</td><td>A. Create CSV file from given data. Read the data from CSV files into a data frame. Perform data pre-processing tasks such as handling missing values and outliers. [20]

<table>
<tr><th>Rollno</th><th>Name</th><th>Age</th><th>Marks</th><th>Class</th></tr>
<tr><td>1</td><td>Sudin</td><td>44</td><td>47</td><td>FY</td></tr>
<tr><td>2</td><td>Shaima</td><td>46</td><td>86</td><td>SY</td></tr>
<tr><td>3</td><td>Raina</td><td>27</td><td>45</td><td>TY</td></tr>
<tr><td>4</td><td>Paul</td><td>38</td><td></td><td>SY</td></tr>
<tr><td>5</td><td>Rahul</td><td>46</td><td>45</td><td>FY</td></tr>
<tr><td>6</td><td>Gopal</td><td></td><td>67</td><td>TY</td></tr>
<tr><td>7</td><td>Yatin</td><td>59</td><td>45</td><td>FY</td></tr>
<tr><td>8</td><td>Jim</td><td>36</td><td>34</td><td>FY</td></tr>
<tr><td>9</td><td>Nima</td><td>45</td><td>32</td><td>TY</td></tr>
</table>

B. Implement Linear Regression on the Iris dataset using Python/R for predicting petal.width on petal.length. [20]</td><td>40</td></tr>
<tr><td></td><td>Viva</td><td>5</td></tr>
<tr><td></td><td>Journal</td><td>5</td></tr>
</table>

SOLUTION – Chit 19

Part A – Data Preprocessing

```
import pandas as pd

data = {
    'Rollno':[1,2,3,4,5,6,7,8,9],
    'Name':['Sudin','Shaima','Raina','Paul','Rahul','Gopal','Yatin','Jim','Nima'],
    'Age':[44,46,27,38,46,None,59,36,45],
    'Marks':[47,86,45,None,45,67,45,34,32],
    'Class':['FY','SY','TY','SY','FY','TY','FY','FY','TY']
}
df = pd.DataFrame(data)
df.to_csv('students.csv', index=False)
df = pd.read_csv('students.csv')

df['Age'].fillna(df['Age'].mean(), inplace=True)
df['Marks'].fillna(df['Marks'].mean(), inplace=True)

Q1 = df['Marks'].quantile(0.25)
Q3 = df['Marks'].quantile(0.75)
IQR = Q3 - Q1
df = df[(df['Marks'] >= Q1 - 1.5*IQR) & (df['Marks'] <= Q3 + 1.5*IQR)]
print(df)
```

Part B – Linear Regression on Iris

```
from sklearn.datasets import load_iris
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
import pandas as pd
import matplotlib.pyplot as plt

iris = load_iris()
df = pd.DataFrame(iris.data, columns=iris.feature_names)

X = df[['petal length (cm)']].values
y = df['petal width (cm)'].values

model = LinearRegression()
model.fit(X, y)
y_pred = model.predict(X)
print("R2:", r2_score(y, y_pred))

plt.scatter(X, y, alpha=0.5, label='Actual')
plt.plot(X, y_pred, color='red', label='Predicted')
plt.xlabel('Petal Length')
plt.ylabel('Petal Width')
plt.title('Linear Regression on Iris Dataset')
plt.legend()
plt.savefig('petal_lr.png')
plt.show()
```

University Of Mumbai

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1.	A. Perform Logistic Regression on the Iris dataset using Python/R to predict binary outcome. Evaluate model's performance using classification metrics. [20] B. Consider the dataset given below. Create Pivot Table in Excel for following analysis and visualize the data using PivotChart: [20] ● Find out the total Sales. ● Find out the Sum of Sales color-wise. ● Find out the Sum of Units. ● Find out Region-wise total sales and total units.	40
	Viva	5
	Journal	5

Date	Color	Region	Units	Sales
03-Jan-20	Red	West	1	110000
14-Jan-20	Blue	South	8	96000
21-Jan-20	Green	West	2	26000
30-Jan-20	Blue	North	7	84000
07-Feb-20	Green	North	8	25000
13-Feb-20	Red	South	2	60000
22-Feb-20	Blue	East	5	35000
01-Mar-20	Green	West	2	87000
13-Mar-20	Blue	East	8	69000
23-Mar-20	Blue	North	7	54000

SOLUTION – Chit 20

Part A – Logistic Regression on Iris

```
from sklearn.datasets import load_iris
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix

iris = load_iris()
X = iris.data[iris.target != 2]
y = iris.target[iris.target != 2]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
clf = LogisticRegression(max_iter=200)
clf.fit(X_train, y_train)
y_pred = clf.predict(X_test)
print(classification_report(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

Part B – Pivot Table Analysis in Python

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    'Date':['03-Jan-20','14-Jan-20','21-Jan-20','30-Jan-20','07-Feb-20',
            '13-Feb-20','22-Feb-20','01-Mar-20','13-Mar-20','23-Mar-20'],
    'Color':['Red','Blue','Green','Blue','Green','Red','Blue','Green','Blue','Blue'],
    'Region':['West','South','West','North','North','South','East','West','East','North'],
    'Units':[1,8,2,7,8,2,5,2,8,7],
    'Sales':[110000,96000,26000,84000,25000,60000,35000,87000,69000,54000]
}
df = pd.DataFrame(data)

print("Total Sales:", df['Sales'].sum())
print("Sales by Color:\n", df.groupby('Color')['Sales'].sum())
print("Total Units:", df['Units'].sum())
print("Region-wise:\n", df.groupby('Region')[['Sales','Units']].sum())

df.groupby('Color')['Sales'].sum().plot(kind='bar', title='Sales by Color', color='skyblue')
plt.tight_layout()
plt.savefig('sales_color.png')
plt.show()

df.groupby('Region')[['Sales','Units']].sum().plot(kind='bar', title='Region-wise Sales & Units')
plt.tight_layout()
plt.savefig('region_sales.png')
plt.show()
```